



Manufacturing Downtime Analysis Project

Tableau | SQL | Python (Pandas) | Data Analysis

This analysis will support data-driven decision-making to enhance production efficiency.

1. Project Overview

This project analyzes production line stoppages to identify critical points affecting operational efficiency. By integrating production, operator, and downtime data, we built interactive dashboards for data-driven decision-making.

2. Objective

The objective of this project is to analyze manufacturing downtime to:

- Identify the main causes of production stoppages
- Reduce downtime duration
- Improve operational & production line efficiency

3. Problem Statement

The manufacturing process experiences frequent downtime events without a clear understanding of their root causes.

This leads to reduced productivity and increased operational costs.

4. Data Description

Dataset 1: Line Downtime

- Batch: Production batch ID
- Downtime Factors (1-12): Causes of downtime
- Downtime: Time lost per event

Dataset 2: Downtime Factors

- Factor: Downtime factor ID
- Description: Explanation of each factor
- Operator Error: Indicates whether the cause is human-related (Yes/No)

Dataset 3: Products

- Product: Product code and size
- Flavor
- Size
- Min Batch Time: Minimum time needed for batch production (with no downtime)

Dataset 4: Line Productivity

- Batch ID: Production batch identifier
- Product: Product code and size
- Operator: Operator name responsible for the batch
- Start DateTime: Batch start date and time
- End DateTime: Batch end date and time
- Duration: Total batch run time calculated from Start and End DateTime

5. Data Preparation

- Cleaning data using PYTHON
- Transform data from wide format to long format using Unpivot
- Ensure correct data types
- Create a relationship between datasets using the “Factor” and “Batch” columns

6. Key Analysis Questions

- What are **the most frequent** downtime causes?
- Which factors **contribute the most** to total downtime?
- What is the **average downtime** per batch?
- Are most downtime events caused by **operator errors or system issues**?
- How do **different shifts** affect production line performance?

7. Key Performance Indicators (KPIs)

- Total Downtime (minutes)
- Average Downtime per event
- Number of Downtime Events
- Percentage of Operator Errors
- Average Cycle Time
- Total Production Output
- Line Availability
- MTBF (Mean Time Between Failures)
- MTTR (Mean Time to Repair)
- Operator with highest downtime rate

8. Dashboard Requirements

- Bar chart showing downtime by factor (Pareto Chart)
- Pie chart showing operator vs non-operator errors (both by no. of incident & Total downtime in min)
- **KPI indicators** (total and average downtime, Total Events, Operator Error %, Non-Operator Error %)
- **Filters** (Batch, Factor, Operator, Product, Date range, operator error (yes/no))
- Downtime trend analysis
- Products with highest downtime rate
- Heatmap Showing the best system of production
- Heatmap showing the weak point for every product
- Operator Downtime Rate for each Operator
- Operational Cost of CO-2L vs. CO-600 ml
- Total impact for each factor

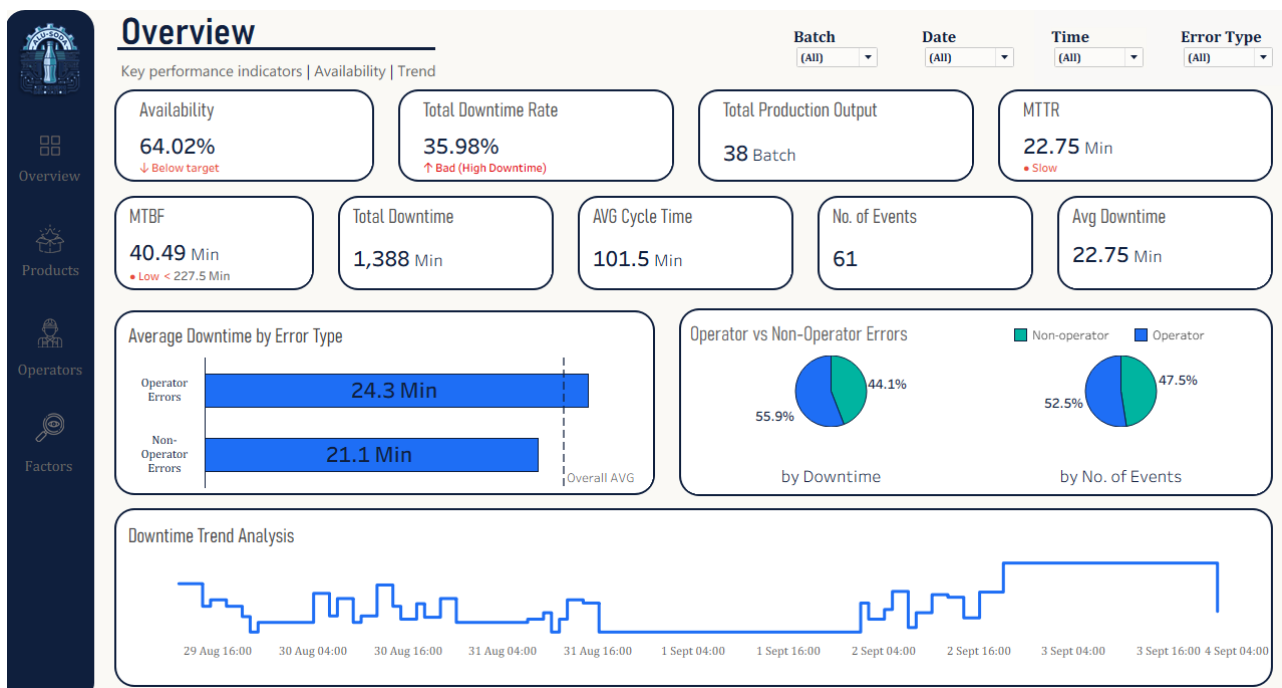
9. Expected Outcomes

- Identification of major downtime causes
- Understanding the impact of human vs machine-related issues
- Actionable insights to reduce downtime

10. Dashboards & Insights

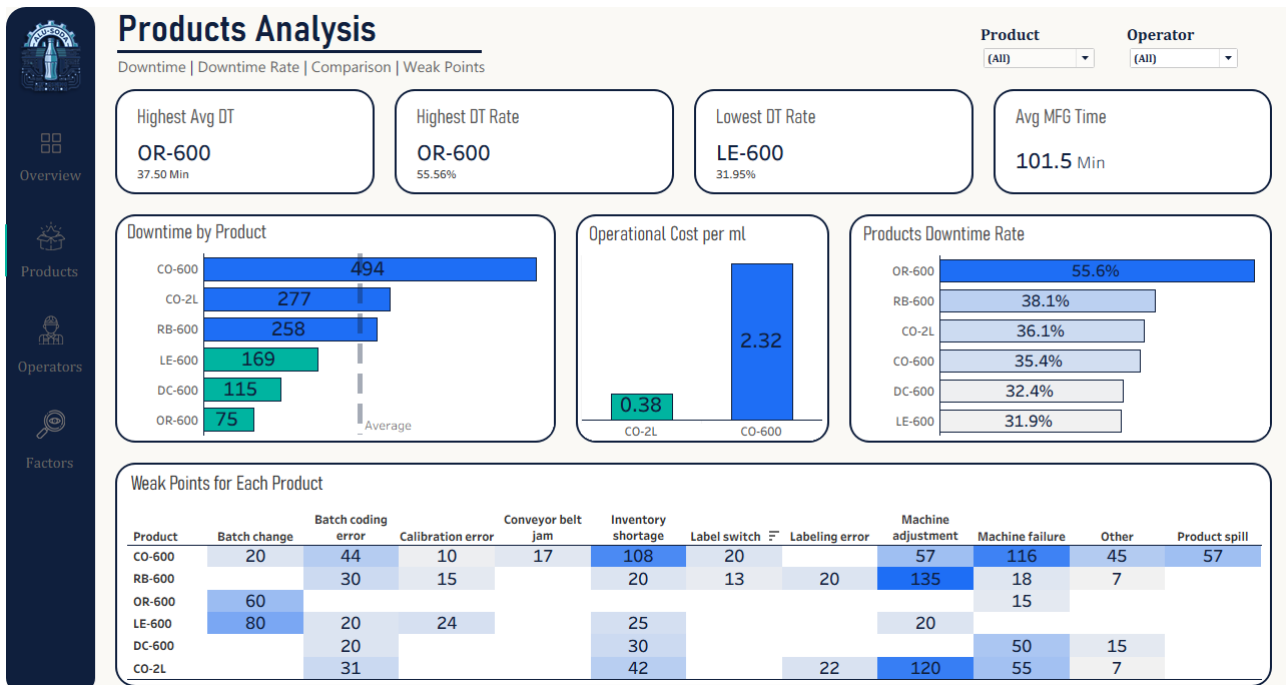
10.1 Operational Overview Dashboard

Focuses on high-level KPIs including Availability, Downtime Rate, MTTR, and MTBF. It tracks the daily production trend and highlights critical peaks in downtime.



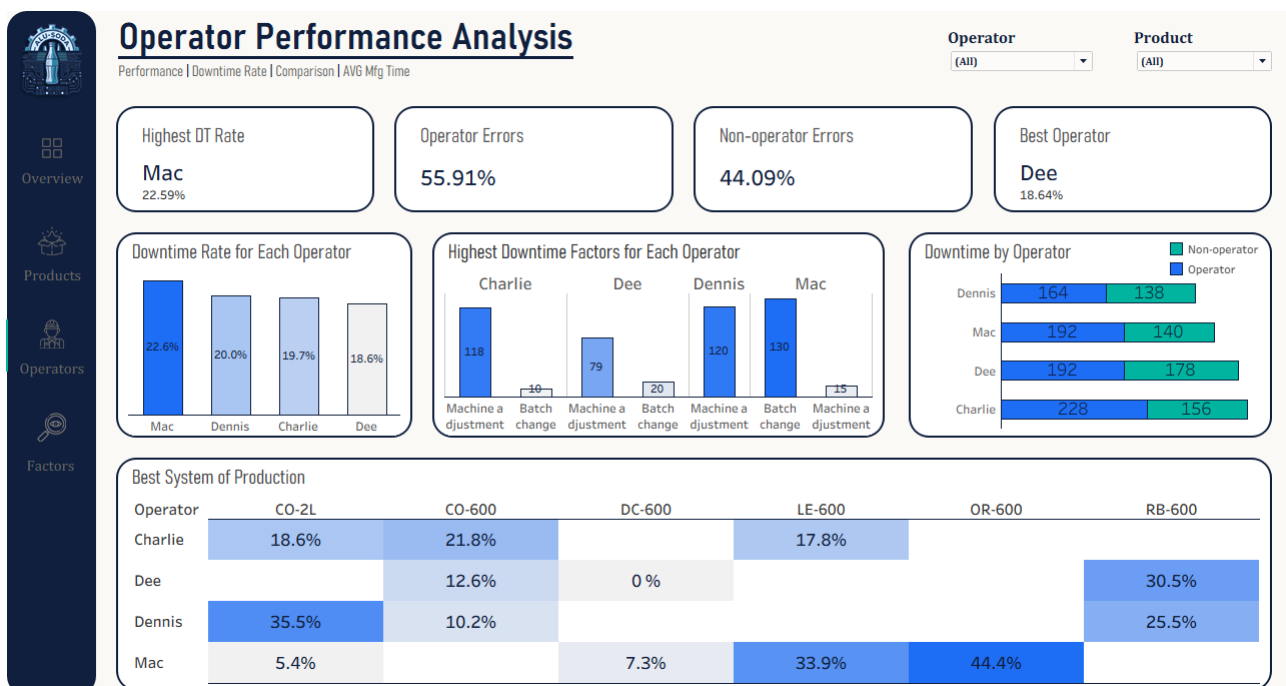
10.2 Products Performance Analysis

Analyzes downtime impact across different product sizes (e.g., CO-600 vs. CO-2L). It highlights which products suffer from the most frequent stoppages and the associated costs



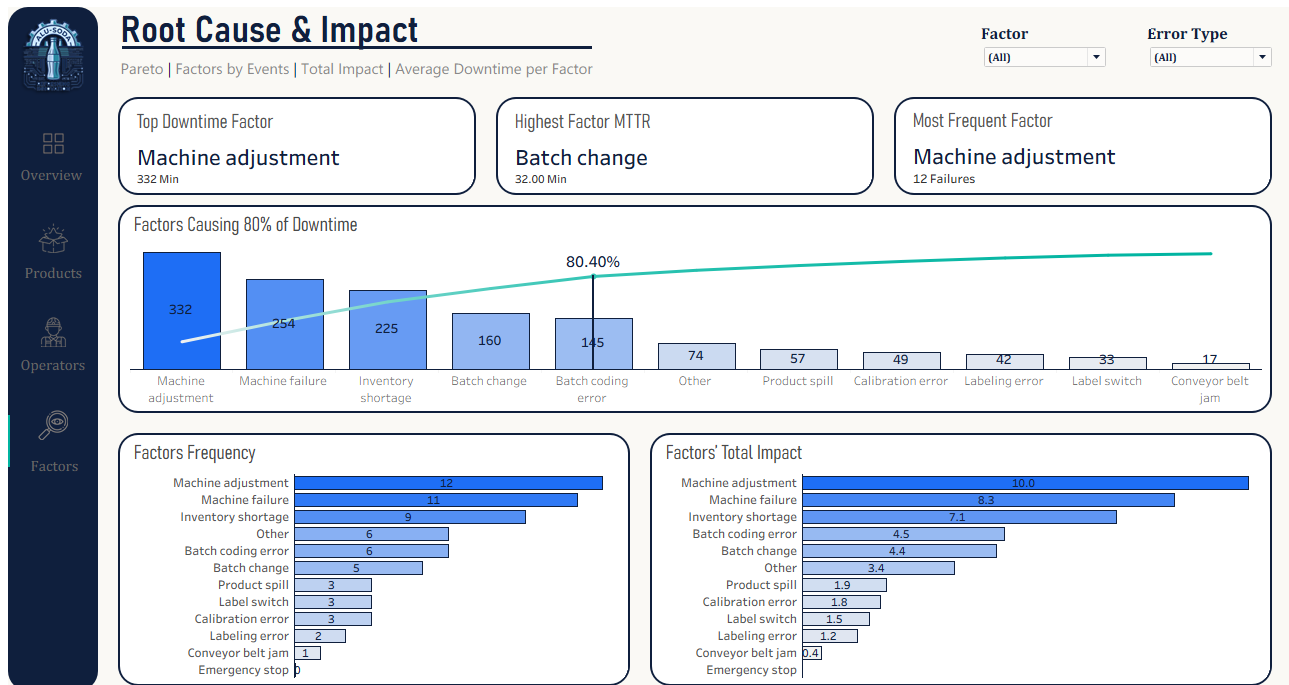
10.3 Operator Performance (Best Systems of Productions)

Evaluates individual operator efficiency and identifies their 'Best Systems of Productions'. It differentiates between machine failure and human-related errors.



10.4 Root Cause (Pareto Analysis)

Utilizes the 80/20 rule to pinpoint the vital few factors causing the majority of downtime. The dashboard highlights the 'Most Frequent Factor' for targeted intervention.



11. Tech Stack

- Python: Data cleaning, unpivoting, and transformation.
- Tableau: Advanced visualization and Elite Dashboard design.

4. Strategic Recommendations

- Implement specialized training programs for operators focusing on “Machine Adjustment” procedures to reduce Operator Errors.
- Re-distribute production loads based on operator-specific efficiency metrics.
- Optimize maintenance schedules for products with high machine-adjustment downtime.
- Improve preventive maintenance strategies.